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IT IS 5250

Forensic Examination in the matter of

The UNC Charlotte stolen-property investigation, Item A thumb drive

Overview

Dr. Robert Quincy sent me this one. Officer Jimmy Johns of the UNC Charlotte Police Department seized a USB thumb drive (Item A) on consent from the subject of a university equipment-theft case, imaged it, and I was asked to go through that image. Dr. Quincy wanted me to report on the following, sticking to what the evidence backs up:

- Any photos relevant to the investigation, including deleted ones.
- Any photos of Apple devices.
- Location info tied to those photos.
- The make and model of the camera that took them.
- Anything in the pictures that helps identify the device or the location.
- Any other pictures whose content might tie back to the theft of university equipment.

Most of what is on the drive is just stuff downloaded off the internet but buried in there were three deleted photos that matter to the case, and I got all three back. They were shot on an iPhone 6s on February 23, 2016, seconds apart in the late afternoon, and each one has GPS baked in that puts it on the UNC Charlotte campus right around Woodward Hall, the building where gear had been going missing and where Officer Johns spotted the subject that day. The photos show University computer and office equipment, and the iPhone 6s that took them lines up with the phone the subject had on him.

One thing is that none of the three photos are of Apple products. They show University owned Dell, Sympodium, and Ricoh gear. The Apple piece here is the camera, the subject's iPhone 6s, not the stuff in the pictures. This is only the Item A drive. The iPhone 6s, and the Apple laptop, iPads, and Mac Mini mentioned in the case file, never came to me, so I did not look at any of those.

Exam Preparation

I got two forensic images of the same thumb drive. Both are 256 MiB E01 images with a single FAT32 partition starting at sector 128, volume labeled JUMPDRIVE. Each one stores its own MD5 and SHA1, and I checked both they verify.

Image file	Case / acquired	MD5 (stored = verified)	Result
ItemA_thumbdrive.E01	X_02232016 — 2016-02-23 22:38	3e7f5fa6e2972014b8671960a0624566	Verified
ItemA_Evidence.E01	X_09172024 — 2024-09-17 18:29	f9ed3e0a1a952dac2fa511e985342e97	Verified

ItemA_thumbdrive.E01 is the one Officer Johns made the day of the stop, so I am treating that as the main evidence. ItemA_Evidence.E01 is a later re image of the same drive. The three photos that matter show up in both, and two of the three come back bit-for-bit identical across the two images, which is what you would expect if it is the same drive.

Accounting for all the space

I also confirmed the image covers the whole device, not just the visible files. Total size = bytes per sector × sector count = $512 \times 524,288 = 268,435,456$ bytes, which is 256 MB. That matches the source size in the acquisition log, so the E01 accounts for every sector of the drive, including the unallocated space where deleted and carved files live.

Tools and workflow

I did the primary analysis in Autopsy 4.23.1 on Windows, the taught workflow: I created the case, added ItemA_thumbdrive.E01 as a disk image, set the case time zone to Eastern, and ran the ingest modules Picture Analyzer/EXIF, Extension Mismatch Detector, PhotoRec Carver, Hash Lookup, File Type Identification, Recent Activity. Autopsy opens the evidence read-only, so nothing was written back to the source. Then I cross checked the key findings with a second, independent toolset, The Sleuth Kit, ExifTool, and ewfverify, and both produced the same results, same recovered files, same iPhone 6s EXIF and GPS, same MD5. Confirming a finding with two tools is the point, the matching hashes and matching EXIF below are that cross-check.

Time zone basis

Different artifacts store time in different zones, so before trusting any date I pinned down the zone. The evidence device was set to Eastern Standard Time. I confirmed this from the photos themselves: IMG_1692's EXIF "Date/Time Original" reads 16:07:59 while its GPS timestamp which is always UTC reads 21:07:50, a 5 hour gap that is UTC-5. February is standard time, so EST, which is correct for Charlotte. I also set the Autopsy case time zone to Eastern, so its columns already read in EST.

Two rules follow, and they matter because the raw numbers do not line up on their own: EXIF "Date/Time Original" is the camera's local time, while the Recycle Bin deletion time comes from a Windows FILETIME, which is stored in UTC. Read straight off the tools, the photos look like they were deleted about six hours after they were taken; converted to a single zone, the gap is about one hour. Every time below is labeled with its zone, and the timeline is normalized to local EST so the sequence is honest.

Analysis

Q1) You should provide any photographs relevant to the investigation.

I recovered three photos that are relevant. In Autopsy they show up under the FAT32 volume's \$RECYCLE.BIN as _MG_1692.JPG, _MG_1693.JPG, and _MG_1694.JPG, each marked with the deleted-file (red X) icon; they also appear as PhotoRec-carved copies under \$CarvedFiles. The leading underscore is FAT overwriting the first character of the name when a file is deleted, so the true names are IMG_1692/1693/1694.JPG. The screenshot below is that Autopsy listing, where the recovered files came from.

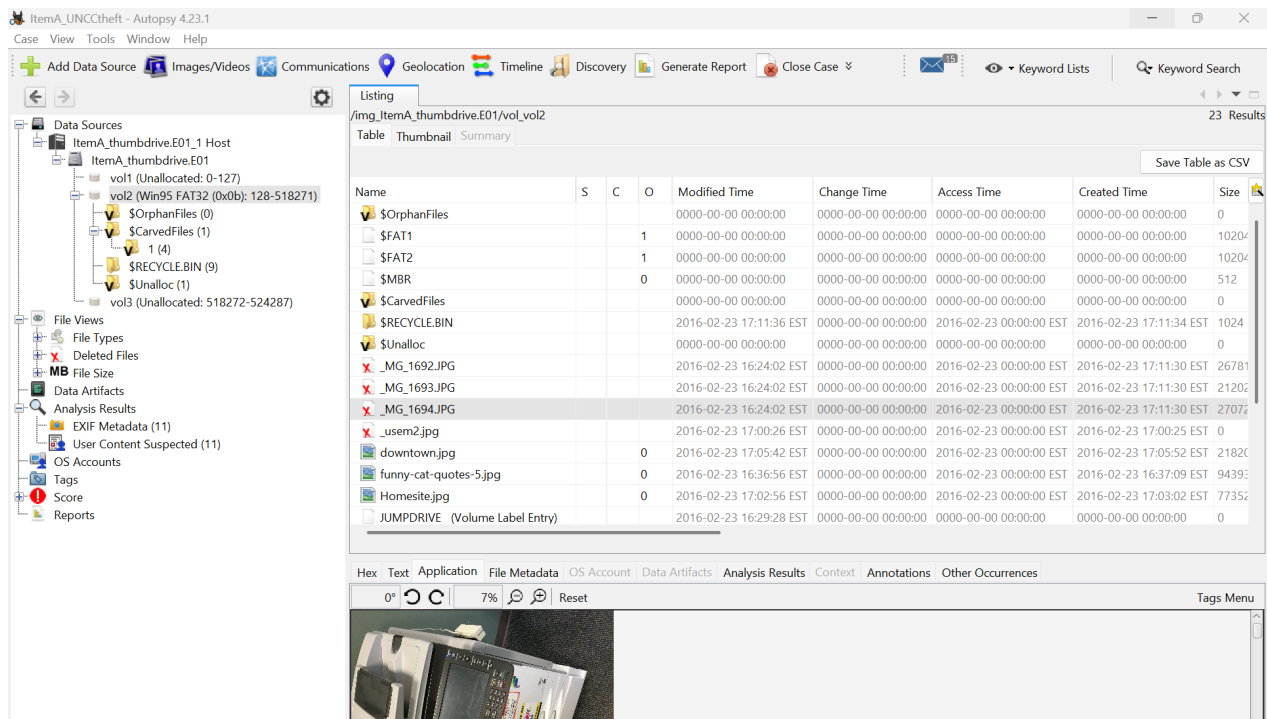


Figure 1: Autopsy

The Recycle Bin also gives the deletion time: 2016-02-23 17:11:36 EST this is a Windows FILETIME, stored as 22:11:36 UTC, about an hour after the photos were taken. Right clicking each photo in Autopsy and choosing Extract File saves the recovered images out for review.

Original file	Taken (EST, local)	GPS (decimal)	What is in the photo
IMG_1692.JPG	2016-02-23 16:07:59 EST	35.307336, -80.736153	Dell OptiPlex 9020 desktop sitting in a wooden AV podium
IMG_1693.JPG	2016-02-23 16:08:11 EST	35.307525, -80.736497	Symposium podium display (Windows 7) on a UNC Charlotte login screen
IMG_1694.JPG	2016-02-23 16:08:30 EST	35.307278, -80.736306	Ricoh Aficio MP C300 "Ink Spot" printer, machine #113



Figure 2: IMG_1692.JPG

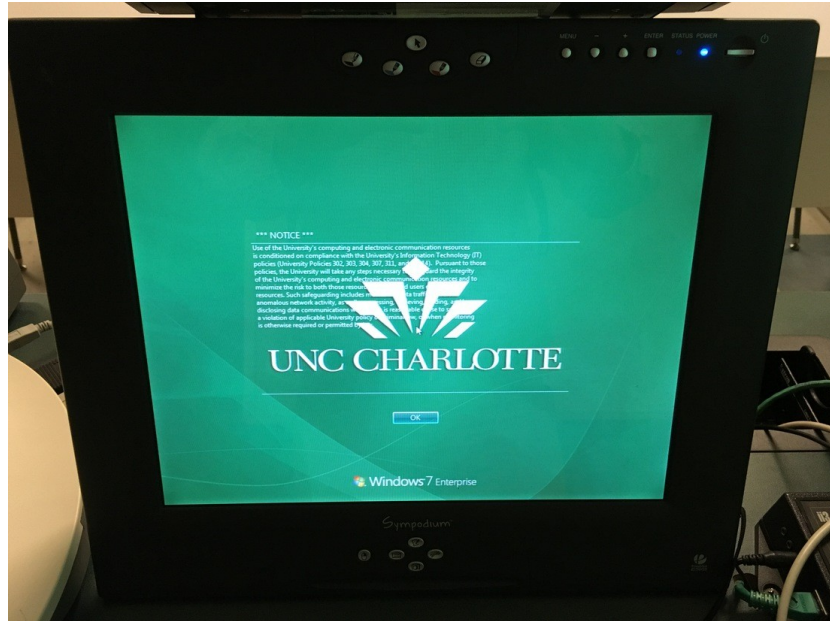


Figure 3: IMG_1693.JPG



Figure 4: IMG_1694.JPG

Q2) Photographs of Apple devices.

None of the recovered photos show an Apple device. The three that matter show no, Apple University gear a Dell, a Symposium display, and a Ricoh printer. The only Apple product this evidence proves is the camera, which I cover in Q4.

Q3) Location information of those devices.

The GPS in the three photos all sits within roughly 35 meters of the same spot on the UNC Charlotte campus, right around Woodward Hall. The coordinates come straight from Autopsy's EXIF Metadata. The pictures back the location up on their own, IMG_1693 shows a podium screen with an UNC CHARLOTTE login notice on it, and IMG_1694 shows a UNCC Ink Spot printing station with a 704-687 University service number. Between the

GPS and what is in the frame, this puts the subject on campus taking pictures of university equipment on the afternoon of February 23, 2016.

Q4) Make/model of the camera device used to make the photos.

All three photos carry the same EXIF pointing at an Apple iPhone 6s. Autopsy's EXIF Metadata result lists Device Make, Apple, Device Model, iPhone 6s, the capture date/time, and the GPS latitude/longitude for each one. That is the same make and model of phone the case file says the subject had in his pocket. The downloaded junk images have different camera info or none, downtown.jpg, for example, reads Nikon D3 / Nikon Corporation from 2012, so it is easy to tell the camera-original photos apart from the stuff pulled off the web. The screenshot below is that EXIF Metadata table.

	Latitude	Longitude	Altitude	Device Model	Device Make	File Path
1 EST	35.307525	-80.73649722222223	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/_MG_1693.JPG
9 EST	35.30733611111111	-80.73615277777778	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/_MG_1692.JPG
0 EST	35.30727777777778	-80.73630555555556	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/_MG_1694.JPG
9 EST	35.30733611111111	-80.73615277777778	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/\$RECYCLE.BIN/
1 EST	35.307525	-80.73649722222223	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/\$RECYCLE.BIN/
0 EST	35.30727777777778	-80.73630555555556	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/\$RECYCLE.BIN/
2 EDT				NIKON D3	NIKON CORPORATION	/img_ItemA_thumbdrive.E01/vol_vol2/downtown.jpg
8 EST	35.30714444444444	-80.73638888888889	205.93290734824282	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/\$CarvedFiles/1..
0 EST	35.30727777777778	-80.73630555555556	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/\$CarvedFiles/1..
1 EST	35.307525	-80.73649722222223	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/\$CarvedFiles/1..
9 EST	35.30733611111111	-80.73615277777778	208.0	iPhone 6s	Apple	/img_ItemA_thumbdrive.E01/vol_vol2/\$CarvedFiles/1..

Figure 5: Autopsy).

Q5) Any content in the picture that might help identify the device.

- IMG_1692: the chassis says Dell OptiPlex 9020, a small-form Dell desktop, with Intel and Windows 7 badges on the front.
- IMG_1693: a Sympodium podium display running Windows 7 Enterprise, sitting on the UNC Charlotte login screen.
- IMG_1694: a Ricoh Aficio MP C300 multifunction printer tagged as a UNCC Ink Spot station, machine number 113, service line 704-687-0809.

The three photos themselves are shown as Figures 2–4 above; each was opened from Autopsy's image viewer.

Q6) Any other pictures whose content might be relevant to the theft of Apple devices.

The rest of the images are downloaded stock photos and junk, not stuff the subject shot. A bunch of them still had Chrome .crdownload leftovers next to them, which tells me they came off the web. That covers Museum.jpg / musem2.jpg, Homesite.jpg, and the meme pictures. One is worth flagging, downtown.jpg is a storefront that reads Central Avenue Jewelry & Pawn. A pawn shop is an obvious place to move stolen gear, so I get why it stands out, but Autopsy's EXIF shows this file is a 2012 Nikon D3 stock photo with no campus GPS, the subject did not take it, so I would treat it as background color, not evidence.



Figure 6: downtown.jpg

Timeline

All times normalized to local Eastern Standard Time (EST, UTC-5). Where a source stores UTC, the original UTC value is shown in parentheses.

- 2016-02-23 4:07:59 – 4:08:30 PM EST — IMG_1692, 1693, 1694 taken on the iPhone 6s at UNC Charlotte, around Woodward Hall. (EXIF local time; GPS confirms 21:07–21:08 UTC.)
- 2016-02-23 5:11:36 PM EST — all three photos sent to the Recycle Bin, about an hour after they were taken. (Recycle Bin FILETIME = 22:11:36 UTC.)
- 2016-02-23 later that evening — Officer Johns images the drive (ItemA_thumbdrive.E01); acquisition metadata records 22:38:43, after the deletion; hashes verify.
- 2024-09-17 6:29 PM — the drive is re-imaged (ItemA_Evidence.E01); hashes verify; the photos are still recoverable.

One objectivity note

both images have an examiner Notes field that says, “This guy is definitely guilty!!” That is an opinion, not a technical note. It does not change anything about the image itself, the hashes still verify but it is worth disclosing, because a defense attorney will absolutely use it to go after how objective the investigation was.

Conclusion

The Item A thumb drive had three deleted photos on it, all recovered clean, all taken with an Apple iPhone 6s on February 23, 2016, and all geotagged by their own EXIF to the UNC Charlotte campus near Woodward Hall which the University branding inside the pictures backs up. They show University computer and office equipment, and they were deleted the same afternoon, about an hour after being taken, and before Officer Johns imaged the drive that evening. The photos themselves are not of Apple products; the Apple device this evidence proves out is the iPhone 6s that took them, which matches the phone the subject was carrying. Autopsy and the command-line toolset returned the same findings, and both images verify against their stored hashes, so I can testify to all of this.

Brian Tramuel
Signed B Tramuel

Post-Examination Hash Re-Verification

After finishing the examination I re verified both images to prove nothing changed while I worked. For each one the MD5 I recomputed matches the MD5 stored at acquisition, so the evidence is intact end to end. This was confirmed three ways, FTK’s acquisition log (MD5 + SHA1, both verified), an independent ewfverify recomputation (matched), and Autopsy, which ingested and ran its Data Source Integrity check against the identical 256 MB image. The screenshot below is the Autopsy data-source properties, showing it loaded the same image (256 MB, 512-byte sectors, Eastern time zone).

Image	MD5 (stored = recomputed)	SHA1 (recomputed)	Result
ItemA_thumbdrive.E01	3e7f5fa6e2972014b8671960a0624566	4c88691b...fd550c	Verified
ItemA_Evidence.E01	f9ed3e0a1a952dac2fa511e985342e97	a1196dc3...1ee6	Verified

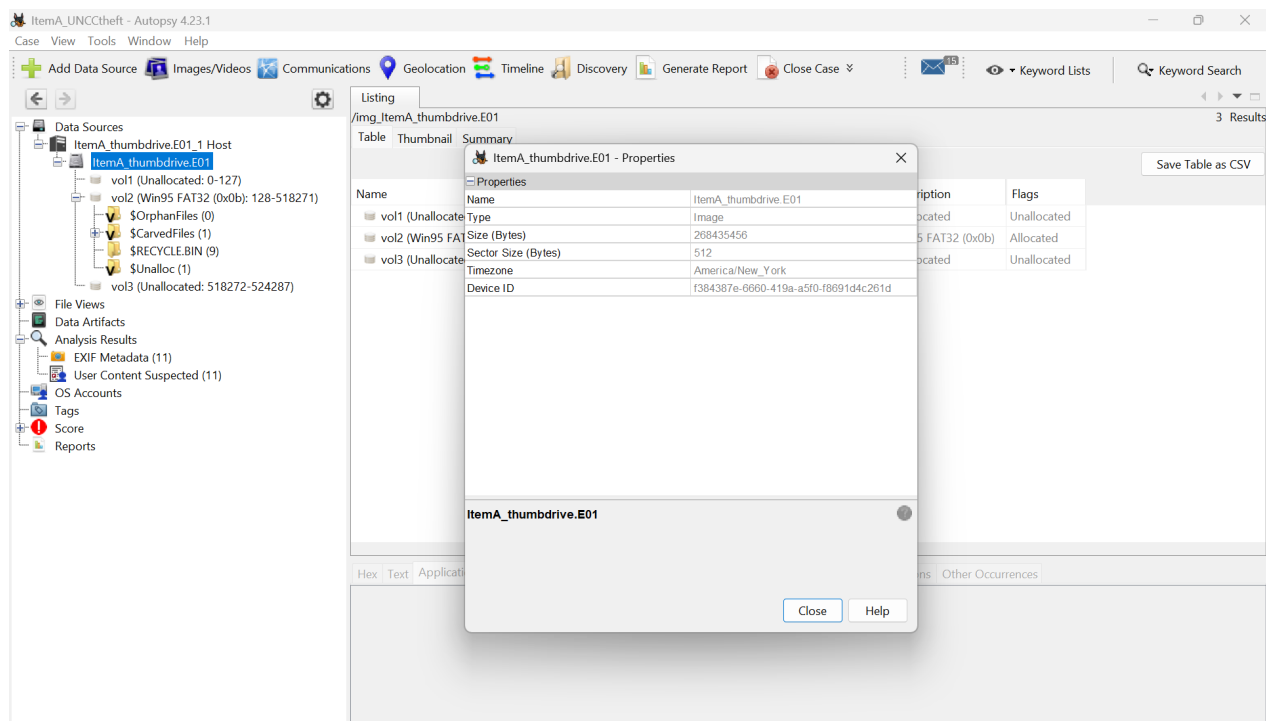


Figure 7: Autopsy

Appendix

Glossary

EnCase / E01 image: A forensic copy of a drive that stores every bit of the original plus its own MD5 and SHA1, so you can prove the copy matches the original.

FAT32: A Microsoft file system you usually see on thumb drives. When a file gets deleted the first letter of the name gets overwritten, which is why IMG_1692 comes back looking like _MG_1692.

EXIF: The metadata a camera stuffs into a photo make, model, timestamp, and GPS if it is turned on.

Recycle Bin \$I / \$R files: When something goes to the Recycle Bin, Windows keeps two files: the \$R with the actual data and the \$I with the original path and the time it was deleted.

.crdownload: A half-finished Chrome download file. If one is sitting next to an image, that image was pulled off the web, not made on the device.

Carving: Pulling files out of unallocated space by scanning for known headers and footers instead of trusting the file system. In Autopsy this is the PhotoRec Carver / \$CarvedFiles.

Full file listing (primary image)

Recovered from ItemA_thumbdrive.E01, FAT32 partition (live and deleted):

- Camera-original (deleted, the ones that matter): IMG_1692.JPG, IMG_1693.JPG, IMG_1694.JPG — also present as Recycle Bin pairs _R/_I FCD13K, F0TJDO, UD9LA4 and as PhotoRec-carved copies.
- Downloaded / decoy images: Homesite.jpg, Museum.jpg (+ .crdownload), musem2.jpg (+ .crdownload), downtown.jpg, science cat.jpg, funny-cat-quotes-5.jpg, onion rings.jpg.

- System: JUMPDRIVE volume label, \$RECYCLE.BIN\desktop.ini.